Chapter 50

TWO-GAP MODEL AND COSTS BENEFITS OF FOREIGN AID

TWO-GAP MODEL

Hollis Chenery and other writers[1] have put forth the "two-gap" approach to economic development. The idea is that "savings gap" and "foreign exchange gap" are two separate and independent constraints on the attainment of a target rate of growth in LDCs. Chenery sees foreign aid as a way of filling these two gaps in order to achieve the target growth rate of the economy.

To calculate the size of gaps, a target growth rate of the economy is postulated along with a given capital-output ratio. A savings gap arises when the domestic savings rate is less than the investment required to achieve the target. For example, if the growth target of national real income is 6 per cent per annum, and the capital-output ratio is 3:1, then the economy must save 18 per cent of its national income to achieve this growth target.[2] If only 12 per cent of savings can be mobilised domestically, the savings gap is 6 per cent of national income. The economy can achieve the target growth rate by filling this savings gap with foreign aid. Similarly, a fixed relationship is postulated between targeted foreign exchange requirements and net export earnings. If net export earnings fall short of foreign exchange requirements, a foreign exchange gap appears which can be filled by foreign aid.

The two gaps are explained in terms of the national income accounting identities:

$$E-Y \equiv I-S \equiv M-X \equiv F$$

where E is national expenditure, Y is national output and income, I is investment, S is saving, M represents imports, X exports and F is net capital inflow.

[1]H. Chenery and A. Strout, "Foreign Assistant and Economic Development," *AER*, Vol. 56, September 1956; H. Chenery and M. Bruno, "Development Alternatives in an Open Economy," *EJ*, Vol. 72, March 1962; and H. Chenery and I. Adelman, "Foreign Aid and Economic Development," *RES*, Vol. 48, Feb. 1966.

[2]This is calculated in terms of Harrod's formula: $Gw = s/Cr$.

$(I—S)$ is the domestic savings gap and $(M—X)$ is the foreign exchange gap. Like the basic national income accounting identities, the two gaps are always equal ex-post in any given accounting period. But they may differ ex-ante because in the long run those who make decisions about savings, investment, exports and imports are different people. So during the planning process, the plans of savers, investors, importers and exporters are likely to be different. Ex-ante (or planned) investment is related to the target growth rate of the economy. If the target growth rate is high, investment will also be high. But domestic savings depend upon the level and distribution of income in the society. Ex-ante imports include the imported inputs needed for development. They are also affected by the size of the national income and the distribution of income among the public and the different sectors of the economy. Exports are exogenously determined by world prices and by quantities that change with weather or natural conditions. As these elements are assumed to be independent of each other, the savings gap and the foreign exchange gap are unequal in size in the ex-ante sense. It is also assumed that savings and foreign exchange cannot be substituted for each other. Further, the country cannot transform its potential savings into exports.

Given these assumptions, Figure 50.1 illustrates the two ex-ante gaps and their relation to different target growth rates of income. The ex-ante savings and foreign exchange gaps, are measured along the vertical axis and the target growth rates along the horizontal axis. The ex-ante savings gap is represented by $(I-S)$ curve and the ex-ante foreign exchange gap by $(M-X)$ curve. Both are equal at point E and the target growth rate of OG is achieved with OF inflow of net foreign aid. If the target growth rate is OG_1, then the foreign exchange gap is larger than the savings gap by ab. This growth rate will not be achieved

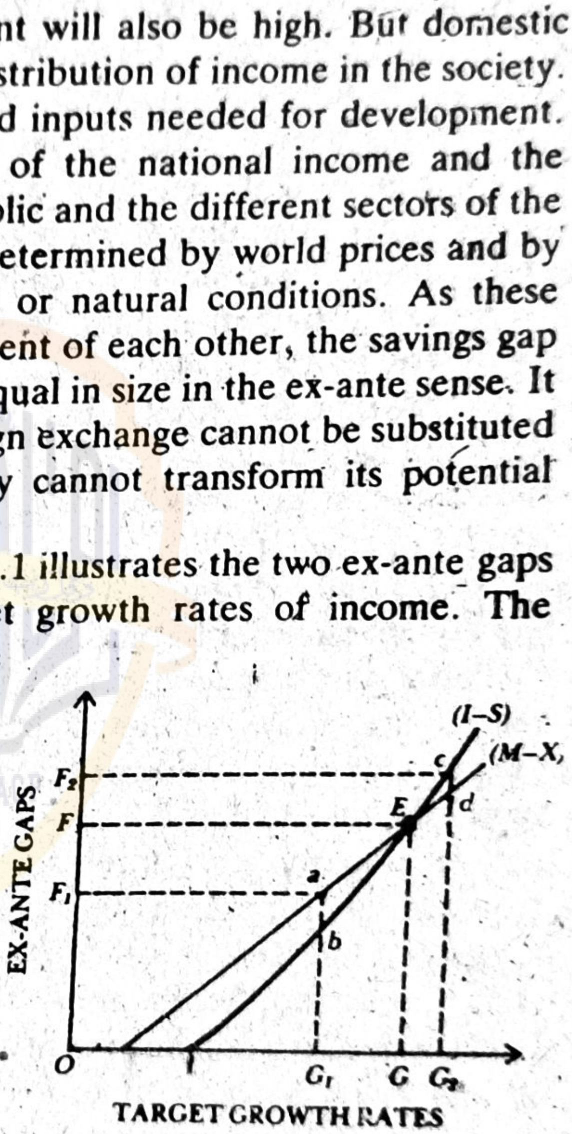

FIG. 50.1

because the inflow of foreign capital is not sufficient to fill the larger foreign exchange gap OF_1. Short run forces might bring about the ex-post equality of the two gaps without achieving the target growth rate. On the other hand, if the target growth rate is OG_2, the savings gap is larger than the foreign exchange gap by cd. Again, this growth rate will not be achieved because the inflow of foreign capital is inadequate to fill the savings gap. It requires a larger inflow of foreign capital to meet the larger savings gap OF_2. Imports cannot be reduced due to "the nature and limited flexibility of the productive system and of the composition of consumer demand." To overcome these structural rigidities, Chenery suggests restrictions on the pattern of consumption, the distribution of income, the level and growth of employment and changes in the exchange rate. Such measures can bring adjustments in the two gaps without foreign aid. But they will retard growth.

Some economists are of the view that if prices are flexible, such rigidities are not likely to be found. If resources are optimally allocated, there can be only savings constraint on growth and hence only savings gap in the economy. If appropriate exchange rate policies or price policies are followed, resources would shift to remove the difference between the growth effect of imports and domestic savings and hence the difference in the ex-ante gaps. This view holds that if the foreign exchange gap is dominant, it must be due to inappropriate price policies which might have led to malallocation of resources.

Assuming that all capital goods are imported and only consumer goods are produced domestically, another two-gap model holds that structural rigidities imply that (*i*) no substitution is possible between imported capital goods and domestic factors in production, and (*ii*) no substitution is possible between different consumer goods in consumption.

The foreign aid required to fill the gap is determined by the dominant gap at a given point in time. If the savings gap is larger than the foreign exchange gap, the economy is said to be in a savings constraint. On the other hand, if the foreign exchange gap is larger than the savings gap, the economy is in a foreign exchange constraint. Foreign aid can help in removing the savings constraint by the inflow of capital. Over the long run, the amount of foreign aid required will equal the difference between the increase in investment and the increase in savings generated by rising income. When the savings gap disappears the target growth rate of the economy will be sustained.

If the foreign exchange constraint is dominant or binding for an LDC at any given point in time, foreign aid can help in overcoming it with foreign aid. The country can start new investment projects by importing capital and intermediate goods and technical assistance. Over the long period, the required foreign aid will equal the difference between the increase in imports and exports. The foreign exchange gap will disappear when exports rise to a level which covers the required imports for the target growth rate of the economy.

Of the two gaps, which dominates first in LDCs, Chenery and Strout cite empirical evidence to show that first such countries have a dominant savings constraint and then the foreign exchange constraint over their course of development. In fact, they divide countries having savings constraint and foreign exchange constraint in two separate categories.

A Critical Appraisal

The two-gap analysis is based on certain restrictive assumptions which limit its usefulness in achieving the target growth rate in LDCs. It presupposes that an increase in domestic savings cannot be utilised as a substitute for the required foreign exchange to maintain investment for the target growth rate. It further assumes that the country cannot follow export promotion and import substitution policies. It also assumes structural rigidities and non-substitutability between different types of goods. Given such rigidities, if the foreign exchange gap is larger than the savings gap, the domestic saving potential can be used neither to produce capital goods nor exports. These assumptions are highly unrealistic and have not been supported by empirical evidence.

Critics point out that the LDCs with dominant savings constraint do not need foreign aid. A dominant savings gap implies that the country is functioning at a full employment level. It is, therefore, not utilising its foreign exchange to import capital goods for investment purposes because the domestic resources are fully employed. As there is full employment, investment in capital goods through imports will lead to inflation.

Moreover, this approach does not consider the absorptive capacity of the economy, and ability to formulate and execute productive projects with aid.

The two-gap analysis is a highly aggregative approach which treats all types of capital investments as homogeneous. This is unrealistic because the capital requirements of LDCs are meant for specific needs and they receive foreign aid for different sectors, industries and projects.

Further, the two gaps are mechanistic. They assume stable values of the parameters in future. But this is unrealistic because the capital-output ratio and the marginal savings rate change over time, depending on domestic conditions and policies. After all, foreign aid cannot be exclusively relied upon to fill these gaps in the long run. With development, structural rigidities are removed and the domestic economy is so transformed as to equilibrate the two gaps. Domestic policies aiming at import substitution and export promotion determine the aid requirements of LDCs. Aid helps in removing rigidities and bringing structural transformation of the economy. Thus the two-gap model is very mechanistic in that it lays emphasis on filling the gaps rather than transforming the economy with aid.

Lastly, the two-gap analysis is at best an approximation for calculating the foreign aid requirements of LDCs. But there are difficulties in fixing import requirements, and in arriving at postulated requirements, requirement, potential domestic savings and expected foreign exchange earnings. It thus provides a crude method of estimating the foreign aid requirements of LDCs.

COSTS AND BENEFITS OF AID

Foreign aid which flows from the donor country in the form of grants, loans, technical assistance, contributions in kind, etc., to the recipient country involves real costs for the former and provides real benefits to the latter. John Pincus[3] has measured the real costs of aid to the donor and the real benefits to the recipient of aid. He measures the real cost of capital flows for a capital exporter as the income he forgoes as a result of the outflow of capital, given alternative possible uses of the same funds. The real benefit for a capital importer is measured by the net increment in income as a result of investing the capital inflow received, as compared with making the same investment with capital from alternative sources.

For measuring capital flows from developed countries (DC) to LDCs, Pincus introduces the concept of a "grant equivalent". A grant is a sort of gift from a DC to an LDC on which the latter is not required to pay any interest or make repayment. A loan given by a DC on soft terms such as lower interest rate, longer grace period and longer repayment period has some concessional element as compared to a loan at commercial market terms. The concessional element in the loan can be converted into its grant equivalent. The concessional element or grant equivalent is the difference between the amount of loan and the present value of repayments discounted at the donor's long-term market rate of interest. Thus the real cost of aid loan to a donor is the difference between the present value of the future repayments discounted at the donor's long-term market rate of interest and the size of the loan.

The present value of an aid loan, therefore, depends upon the interest rate charges relative to the rate of return (discount rate) earned by the donor if the same amount had been invested at home, and on the period of repayment of the aid loan. If the loan rate equals the rate of return in the donor country, then the grant equivalent is zero and the loan is costless to the lender. If the rate of return in the donor country is higher than the loan rate, the real cost of the loan will be higher for the lender. On the contrary, if the loan rate is higher than the rate of return, the real cost will be less to the lender who will gain by lending.

The real benefit of aid loan to the recipient may differ from its real

[3]John Pincus, *Economic Aid and International Cost-sharing*, 1965, and *Costs and Benefits of Aid: An Empirical Analysis*, UNCTAD, 1967.

cost to the donor. The real benefit of a loan will depend upon the rate of return (discount rate) in the recipient country relative to the interest rate charged by the donor. If the rate of return on a similar loan in the recipient country is higher than the interest rate on aid loan, the real benefit of a loan to the borrower will be greater, and *vice versa*.

Besides the rate of return and the loan rate, the real cost and the real benefit of aid loan depends upon the period of grace and period of repayment. If the rate of return is higher than the loan rate in the lender country and the grace period and the repayment period are longer, the real cost to the donor will be high, and *vice versa*. On the other hand, if the rate of return in the recipient country is higher than the loan rate and the grace and repayment periods are longer, the real benefit to the donor will be larger, and *vice versa*.

However, it is difficult to calculate the real cost of aid when it is tied and is in the form of contributions in kind. If the donor ties the aid by requiring the recipient to import from the donor, the grant equivalent will be reduced for the recipient. The real cost to the donor may be less because it may be supplying its goods at prices higher than world market prices. Since the grant equivalent is reduced in this case, the real benefit to the recipient of tied aid is also less. Similar is the case when the donor gives aid in the form of contributions in kind such as surplus agricultural commodities valued at prices higher than world market prices. The opposite will be the case when aid is not tied and goods are valued at world market prices. Therefore, in computing the real cost of foreign aid to donors in a fully employed economy, John Pincus includes the sum of (*i*) grants, including technical assistance, at nominal value; (*ii*) loans valued at the difference between the amount of loan and the present value discounted at the market rate of interest; (*iii*) contributions in kind valued at world market prices; and (*iv*) sales of goods or loans repayable in recipient country's inconvertible currency valued as grants, after making allowance for funds actually spent by the donor in the recipient country.

This analysis of computation of the real cost to the donor is based on certain *assumptions*. First, it is assumed that resources are fully employed in the donor country. In case resources are underemployed in such a country, aid involves a real cost if resources are shifted from actual domestic projects to foreign aid. If the domestic projects have not been actually adopted, then the aid does not involve real cost. Second, this analysis assumes that prices of goods and services under tied aid are valued at world market prices. Third, the grant equivalent for each year is measured without deducting the debt service payments of earlier loans.

The above analysis can also be applied to the real benefit of capital inflows for recipients. The higher the grant equivalent to recipient, the greater will be the real benefit to them. The grant equivalent is higher if the terms of aid are lenient. Similarly if the major portion of the aid is non-tied, the greater is the real benefit to the recipient country. Under such a situation, the same nominal amount of capital inflow will lead to the increase in the real benefit. If the conditions of aid are made softer and the adverse effects of aid-tying are reduced, the real benefit of capital inflow can more closely approach the nominal value of the inflow of capital to the recipient country.

Though the real benefits of aid are measured like its real costs, yet its values normally diverge from the latter due to a number of reasons. First, the interest rate for discounting usually differs in donor and recipient countries. If the discount rate in the recipient country is lower than that in the donor country, the grant equivalent to the recipient is less as compared to the donor. Second, in countries which adopt exchange controls the discount rate to be paid by the recipient country will equal the rate in the international securities market. If the donor country's currency is over-valued the discount rate of the recipient country would have to be increased accordingly. Third, the tied aid will reduce the grant equivalent if the donor country charges higher prices than the world prices for its goods and services. This will lead to a divergence between donor's cost and recipient's benefit estimates of grant equivalent. Fourth, it is difficult to calculate recipient's benefit from private investment. In particular, the beneficial effects of technical assistance and technological transfer to the recipient country are beset with many practical difficulties. All these considerations lead to the divergence of real benefits of aid to the recipient countries from the real costs to the donor countries and thereby tend to reduce the estimates of grant equivalents for recipients and donors.

Its Implications

The cost-benefit analysis of foreign aid leads to certain policy implications. Non-tied aid increases the real cost to the donor country. Tied aid reduces the real cost to the donor. At the same time, tied aid increases the burden of repayment for the recipient and reduces the real benefit of the aid. It is, therefore, advisable that the recipient should insist on non-tied aid as far as feasible.

This analysis lays emphasis on the DCs to ease the terms and conditions of aid to the LDCs. They should so adjust the aid that the grant equivalent per unit value of aid given should increase rather than decline. This argument gains greater force from the fact that a number of LDCs have very high debt service obligations and they find it difficult to repay their accumulated debts. Debt service in such countries competes with essential imports for foreign exchange earnings, thereby adversely affecting domestic savings, investment and hence development. Therefore, the DCs should raise the grant equivalent per dollar of foreign aid to the LDCs. It is better to charge a rate of interest lower than the rate of return on loan in the recipient country.

Further, there is always the fear that the recipient may default on loan repayment. Therefore, the donor should provide grants rather than loans. But due to psychological and political reasons, it is in the interest of the recipient country to provide it soft loans rather than grants.